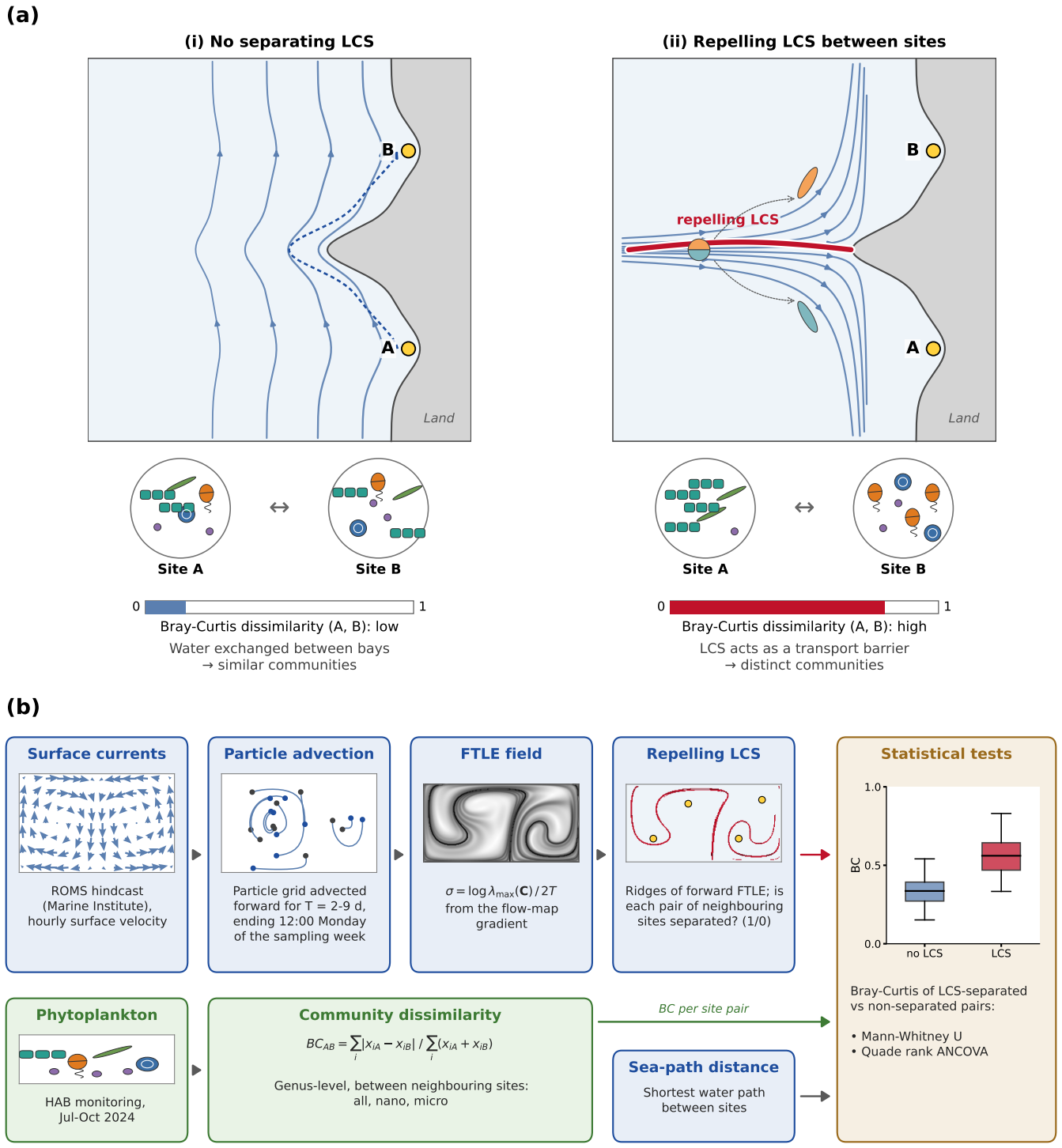

(a)
(i) No separating LCS
(ii) Repelling LCS between sites
B
A
Land
repelling LCS
Site A
Site B
0
1
Bray-Curtis dissimilarity (A, B): low
Water exchanged between bays
→ similar communities
Bray-Curtis dissimilarity (A, B): high
LCS acts as a transport barrier
→ distinct communities
(b)
Surface currents
ROMS hindcast
(Marine Institute),
hourly surface velocity
Particle advection
Particle grid advected
forward for T = 2-9 d,
ending 12:00 Monday
of the sampling week
FTLE field
$\sigma = \log \lambda_{max}(\mathbf{C}) / 2T$
from the flow-map
gradient
Repelling LCS
Ridges of forward FTLE; is
each pair of neighbouring
sites separated? (1/0)
Statistical tests
BC
1.0
0.5
0.0
no LCS
LCS
Bray-Curtis of LCS-separated
vs non-separated pairs:
• Mann-Whitney U
• Quade rank ANCOVA
Phytoplankton
HAB monitoring,
Jul-Oct 2024
Community dissimilarity
$BC_{AB} = \sum_i |x_{iA} - x_{iB}| / \sum_i (x_{iA} + x_{iB})$
Genus-level, between neighbouring sites:
all, nano, micro
BC per site pair
Sea-path distance
Shortest water path
between sites